

Learners have to come up with a Report to support the answers to the following questions and suggestions

Objective Questions

1. Does any table have missing values or duplicates? If yes how would you handle it ?
2. Find the top-selling tracks and top artist in the USA and identify their most famous genres.
3. What is the customer demographic breakdown (age, gender, location) of Chinook's customer base?
4. Calculate the total revenue and number of invoices for each country, state, and city:
5. Find the top 5 customers by total revenue in each country
6. Identify the top-selling track for each customer
7. Are there any patterns or trends in customer purchasing behavior (e.g., frequency of purchases, preferred payment methods, average order value)?
8. What is the customer churn rate?
9. Calculate the percentage of total sales contributed by each genre in the USA and identify the best-selling genres and artists.
10. Find customers who have purchased tracks from at least 3 different genres
11. Rank genres based on their sales performance in the USA
12. Identify customers who have not made a purchase in the last 3 months

Subjective Questions

1. Recommend the three albums from the new record label that should be prioritised for advertising and promotion in the USA based on genre sales analysis.
2. Determine the top-selling genres in countries other than the USA and identify any commonalities or differences.
3. Customer Purchasing Behavior Analysis: How do the purchasing habits (frequency, basket size, spending amount) of long-term customers differ from those of new customers? What insights can these patterns provide about customer loyalty and retention strategies?
4. Product Affinity Analysis: Which music genres, artists, or albums are frequently purchased together by customers? How can this information guide product recommendations and cross-selling initiatives?
5. Regional Market Analysis: Do customer purchasing behaviors and churn rates vary across different geographic regions or store locations? How might these correlate with local demographic or economic factors?
6. Customer Risk Profiling: Based on customer profiles (age, gender, location, purchase history), which customer segments are more likely to churn or pose a higher risk of reduced spending? What factors contribute to this risk?
7. Customer Lifetime Value Modeling: How can you leverage customer data (tenure, purchase history, engagement) to predict the lifetime value of different customer segments? This could inform targeted marketing and loyalty program strategies. Can you observe any common characteristics or purchase patterns among customers who have stopped purchasing?
8. If data on promotional campaigns (discounts, events, email marketing) is available, how could you measure their impact on customer acquisition, retention, and overall sales?
9. How would you approach this problem, if the objective and subjective questions weren't given?
10. How can you alter the "Albums" table to add a new column named "ReleaseYear" of type INTEGER to store the release year of each album?

11. Chinook is interested in understanding the purchasing behavior of customers based on their geographical location. They want to know the average total amount spent by customers from each country, along with the number of customers and the average number of tracks purchased per customer. Write an SQL query to provide this information.